

PRINCIPLES OF GEOGRAPHIC INFORMATION SYSTEM SEM : VI

SEM VI : UNIT 3



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1. What are the different classifications of Map Projections? Explain any two. (Apr-19)

OR

How Map projection are classified? Explain (Apr-23)

Ans:

Maps are one of the world's oldest types of document. For quite some time it was thought that our planet was flat, and during those days, a map simply was a miniature representation of a part of the world. Now that we know that the Earth's surface is curved in a specific way, we know that a map is in fact a flattened representation of some part of the planet. The field of map projections concerns itself with the ways of translating the curved surface of the Earth into a flat map.

A map projection is a mathematically described technique of how to represent the Earth's curved surface on a flat map.

Classification of map projections:

Types of Map Projections Map projections can be grouped by either the preserved property or the projection surface.

Cartographers group map projections by the preserved property into the following four classes: conformal, equal area or equivalent, equidistant, and azimuthal or true direction.

A conformal projection preserves local angles and shapes. An equivalent projection represents areas in correct relative size.

An equidistant projection maintains consistency of scale along certain lines. And an azimuthal projection retains certain accurate directions.

The preserved property of a map projection is often included in its name such as the Lambert conformal conic projection or the Albers equal area conic projection.

The conformal and equivalent properties are mutually exclusive. Otherwise a map projection can have more than one preserved property, such as conformal and azimuthal.

The conformal and equivalent properties are global properties, meaning that they apply to the entire map projection. The equidistant and azimuthal properties are local properties and may be true only from or to the center of the map projection.

The preserved property is important for selecting an appropriate map projection for thematic mapping. For example, a population map of the world should be based on an equivalent projection. By representing areas in correct size, the population map can create a correct.



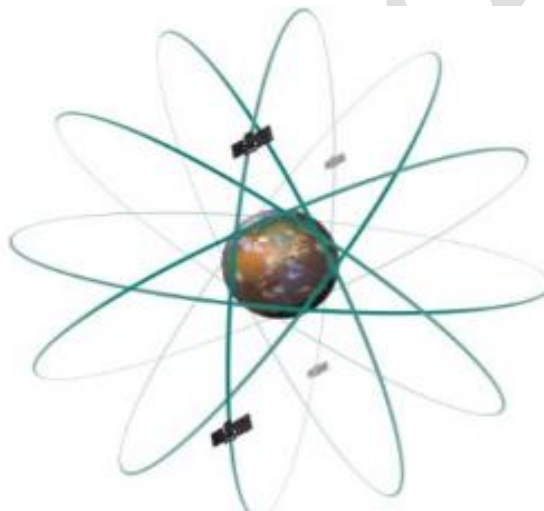
2. Write a note on GPS.

(Apr-19)

Ans:

The NAVSTAR Global Positioning System (GPS) was declared operational in 1994, providing Precise Positioning Services (PPS) to US and allied military forces as well as US government agencies, and Standard Positioning Services (SPS) to civilians throughout the world. Its space segment nominally consists of 24 satellites, each of which orbit our planet in 11h58m at an altitude of 20,200 km. There can be any number of satellites active, typically between 21 and 27. The satellites are organized in six orbital planes, somewhat irregularly spaced, with an angle of inclination of 55–63° with the equatorial plane, nominally having four satellites each.

This means that a receiver on Earth will have between five and eight (sometimes up to twelve) satellites in view at any point in time. Software packages exist to help in planning GPS surveys, identifying expected satellite set-up for any location and time. GPS's control segment has its master control in Colorado, US, and monitor stations in a belt around the equator, namely in Hawaii, Kwajalein Atoll in the Marshall Islands, Diego Garcia (British Indian Ocean Territory) and Ascension Island (UK, southern Atlantic Ocean).



The NAVSTAR satellites transmit two radio signals, namely the L1 frequency at 1575.42 MHz and the L2 frequency at 1227.60 MHz. There are also a third and fourth signal, but they are not important for our discussion here. The first two signals consist of:

- The carrier waves at the given frequencies,
- A coarse ranging code, known as C/A, modulated on L1,
- An encrypted precision ranging code, known as P(Y), modulated on L1 and L2, and
- A navigation message modulated on both L1 and L2.

The role of L2 is to provide a second radio signal, thereby allowing (the more expensive) dual-

frequency receivers a way of determining fairly precisely the actual ionospheric delay on satellite signals received. The role of the ranging codes is twofold:

- 1 To identify the satellite that sent the signal, as each satellite sends unique codes, and the receiver has a look-up table for these codes, and
- 2 To determine the signal transit time, and thus the satellite's pseudorange.

The navigation message contains the satellite orbit and satellite clock error information, as well as some general system information. GPS also carries a fifth, encrypted military signal carrying the M-code. GPS uses WGS84 as its reference system. It has been refined on several occasions and is now aligned with the ITRF at the level of a few centimetres worldwide. GPS has adopted UTC as its time system.

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3. Explain 2D geographic coordinate system using suitable example. (Apr-19)

OR

Explain 2D geographic coordinate system using suitable example.(Nov-19)

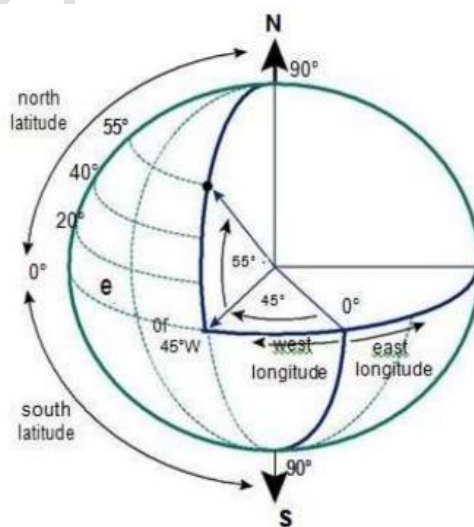
OR

Explain 2D geographic coordinate system.

(Apr-23)

Ans:

- The most widely used global coordinate system consists of lines of geographic latitude (ϕ or $^\circ$) and longitude (λ or $^\circ$). Lines of equal latitude are called parallels. They form circles on the surface of the ellipsoid. Lines of equal longitude are called meridians and they form ellipses (meridian ellipses) on the ellipsoid.
- Latitude is zero on the equator ($= 0^\circ$), and increases towards the two poles to maximum values of $= +90^\circ$ (N 90°) at the North Pole and $= -90^\circ$ (S 90°) at the South Pole.
- The longitude ($^\circ$) is the angle between the meridian ellipse which passes through Greenwich and the meridian ellipse containing the point in question. It is measured in the equatorial plane from the meridian of Greenwich ($= 0^\circ$) either eastwards through $= +180^\circ$ (E 180°) or westwards through $= -180^\circ$ (W 180°)
- Latitude and longitude represent the geographic coordinates ($^\circ$; $^\circ$) of a point P' (Figure 10) with respect to the selected reference surface. They are always given in angular units. For example, the coordinates for City hall in Enschede are:
 $= 52^\circ 13' 02.6''$ N; $= 6^\circ 53' 03.2''$ E
- The graticule on a map represents the projected position of the geographic coordinates ($^\circ$; $^\circ$) at constant intervals, or in other words the projected position of selected meridians and parallels. The shape of the graticule depends largely on the characteristics of the map projection and the scale of the map.



4. What is trend surface fitting? Explain.

(Apr-19)

Ans:

Since the data are continuous, we can make use of measured values for inter-polation. There are many continuous geographic fields—elevation, temperature and ground water salinity are just a few examples. Commonly, continuous fields are represented as rasters, and we will almost by default assume that they are. Alternatives exist though. The main alternative for continuous field representation is a poly line vector layer in which the lines are isolines. We will also address these issues of representation below.

The aim is to use measurements to obtain a representation of the entire field using point samples. In this section we outline four techniques to do so:

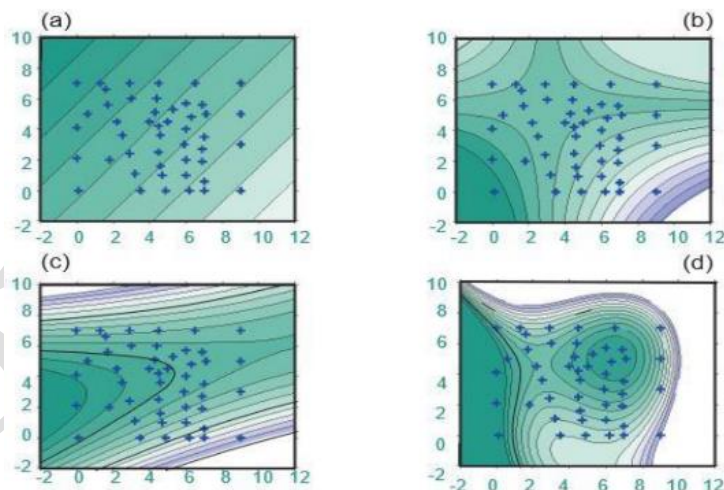
1. Trend surface fitting using regression,
2. Triangulation,
3. Spatial moving averages using inverse distance weighting
4. Kriging.

- **Trend surface fitting:**

In trend surface fitting, the assumption is that the entire study area can be represented by a formula $f(x, y)$ that for a given location with coordinates (x, y) will give us the approximated value of the field in that location.

The key objective in trend surface fitting is to derive a formula that best describes the field. Various classes of formula exist, with the simplest being the one that describes a flat, but tilted plane:

$$f(x, y) = c_1 \cdot x + c_2 \cdot y + c_3.$$



If we believe—and this judgement must be based on domain expertise—that the field under consideration can be best approximated by a tilted plane, then the problem of finding the best plane is the problem of determining best values for the coefficients c_1 , c_2 and c_3 . This is where the point measurements earlier obtained become important. Statistical techniques known as regression techniques can be used to determine values for these coefficients c_i that best fit with the measurements. A plane will be fitted through the measurements that makes the smallest overall error with respect to the original measurements.

In Figure 5.15, we have used the same set of point measurements, with four different approximation functions. Part (a) has been determined under the assumption that the field can be approximated by a tilted plane, in this case with a downward slope to the southeast. The values found by regression techniques were:

$c_1 = -1.83934$, $c_2 = 1.61645$ and $c_3 = 70.8782$, giving us:

$$f(x, y) = -1.83934 \cdot x + 1.61645 \cdot y + 70.8782.$$

Clearly, not all fields are representable as simple, tilted planes. Sometimes, the theory of the application domain will dictate that the best approximation of the field is a more complicated, higher-order polynomial function. Three such functions were the basis for the fields illustrated in Figure 5.15(b)–(d).

The simplest extension from a tilted plane, that of bilinear saddle, expresses some dependency between the x and y dimensions:

$$f(x, y) = c_1 \cdot x + c_2 \cdot y + c_3 \cdot xy + c_4.$$

This is illustrated in part (b). A further step up the ladder of complexity is to consider quadratic surfaces, described by:

$$f(x, y) = c_1 \cdot x^2 + c_2 \cdot x + c_3 \cdot y^2 + c_4 \cdot y + c_5 \cdot xy + c_6.$$

The objective is to find six values for our coefficients that best match with the measurements. A bilinear saddle and a quadratic surface have been fitted through our measurements in Figure 5.15(b) and (c), respectively.

Part (d) of the figure illustrates the most complex formula of the surfaces in Figure 5.15, the cubic surface. It is characterized by the following formula:

$$f(x, y) = c_1 \cdot x^3 + c_2 \cdot x^2 + c_3 \cdot x + c_4 \cdot y^3 + c_5 \cdot y^2 + c_6 \cdot y + c_7 \cdot x^2 y + c_8 \cdot xy^2 + c_9 \cdot xy + c_{10}.$$

Trend surface fitting is a useful technique of continuous field approximation, though determining the ‘best fit’ values for the coefficients c_i is a time-consuming operation, especially with many point measurements. Once these best values have been determined, we know the formula, making it possible to compute an approximated value for any location in the study area.

It is possible to use trend surfaces for both global and local trends. Global trend surface fitting is based on the assumption that the entire study area can be approximated by the same mathematical surface. However in many cases, the assumption that a single formula can describe the field for the entire study area is an unrealistic one. Capturing all the fluctuation of a natural geographic field in a reasonably sized study area, demands polynomials of extreme orders, and these quickly become computationally impossible to decipher.

5. How Root Mean Square is used to mean location accuracy? Explain.

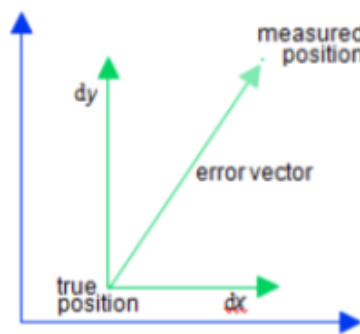
(Apr-19)

Ans:

- **Root mean square error:**

Location accuracy is normally measured as a root mean square error (RMSE). The RMSE is similar to, but not to be confused with, the standard deviation of a statistical sample. The value of the RMSE is normally calculated from a set of check measurements (coordinate values from an independent source of higher accuracy for identical points). The differences at each point can be plotted as error vectors, as is done in Figure 5.3 for a single measurement. The error vector can be seen as having constituents in the x- and y directions which can be recombined by vector addition to give the error vector representing its locational error.

For each checkpoint, the error vector has components δx and δy . The observed errors should be checked for a systematic error component, which may indicate a (possibly repairable) lapse in the measurement method. Systematic error has occurred when $\delta x = 0$ or $\delta y = 0$.



The systematic error δx in x is then defined as the average deviation from the true value:

$$\delta x = \frac{1}{n} \sum_{i=1}^n \delta x_i.$$

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6. Write a note on Krigging.**(Apr-19)****Ans:****• Krigging:**

- Kriging was originally developed by mining geologists attempting to derive accurate estimates of mineral deposits in a given area from limited sample measurements. It is an advanced interpolation technique belonging to the field of geostatistics, which can deliver good results if applied properly and with enough sample points.
- Kriging is usually used when the variation of an attribute and/or the density of sample points is such that simple methods of interpolation may give unreliable predictions.
- Kriging is based on the notion that the spatial change of a variable can be described as a function of the distance between points. It is similar to IDW interpolation, in that it the surrounding values are weighted to derive a value for an unmeasured location.
- However, the kriging method also looks at the overall spatial arrangement of the measured points and the spatial correlation between their values, to derive values for an unmeasured location.
- The first step in the kriging procedure is to compare successive pairs of point measurements to generate a semi-variogram.
- In the second step, the semi-variogram is used to calculate the weights used in interpolation.
- Although kriging is a powerful technique, it should not be applied without a good understanding of geostatistics, including the principle of spatial autocorrelation.
- Kriging is a multistep process; it includes exploratory statistical analysis of the data, variogram modeling, creating the surface, and (optionally) exploring a variance surface.
- Kriging is most appropriate when you know there is a spatially correlated distance or directional bias in the data.
- Kriging predicts the value of a function at a given point by computing a weighted average of the known values of the function in the neighborhood of the point.
- The method is closely related to regression analysis.

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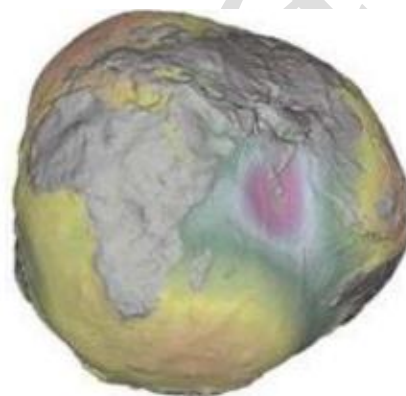
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7. Write a note on the Geoid and vertical datum.**(Nov-19)****Ans:****• The Geoid:**

- Imagining that the entire Earth's surface is covered by water, if we ignore tidal and current effects on this 'global ocean', the resultant water surface is affected only by gravity.
- This has an effect on the shape of this surface because the direction of gravity—more commonly known as plumb line is dependent on the mass distribution inside the Earth.
- Due to irregularities or mass anomalies in this distribution the 'global ocean' results in an undulated surface. This surface is called the Geoid (Figure). The plumb line through any surface point is always perpendicular to it.
- The Geoid is used to describe heights. In order to establish the Geoid as reference for heights, the ocean's water level is registered at coastal places over several years using tide gauges (mareographs).
- Averaging the registrations largely eliminates variations of the sea level with time. The resulting water level represents an approximation to the Geoid and is called the mean sea level.

**• Vertical datum:**

- There are several realizations of local mean sea levels (also called local vertical datums) in the world. They are parallel to the Geoid but offset by up to a couple of meters.
- This offset is due to local phenomena such as ocean currents, tides, coastal winds, water temperature and salinity at the location of the tide gauge.
- Care must be taken when using heights from another local vertical datum.
- The local vertical datum is implemented through a levelling network.
- A levelling network consists of benchmarks, whose height above mean sea level has been determined through geodetic levelling. The implementation of the datum enables easy user access.
- The surveyors do not need to start from scratch (i.e. from the Amsterdam tide-gauge) every time they need to determine the height of a new point. They can use the benchmark of the levelling network that is closest to the point of interest.

8. What is secondary data in GIS? Explain any two ways to obtain secondary data in GIS. (Nov-19)

Ans:

In GIS, secondary data refers to information that has been collected, processed, and stored by someone else for a purpose other than the one at hand. This data is not originally created for the specific GIS project but is utilized for analysis, mapping, or other geographic applications. Obtaining secondary data is a common practice in GIS to leverage existing information.

Spatial data can also be sourced indirectly. This includes data derived from existing paper maps. Secondary data through scanning, data digitized from a satellite image, processed data purchased from data capture firms or international agencies, and so on. This type of data is known as secondary data. Here are explanations of two ways to obtain secondary data in GIS: digitizing and scanning.

- **Two ways to obtain secondary data in GIS:**

- **Digitizing**

- A traditional method data is through digitizing existing paper maps. This can be done using various techniques. Before adopting this approach, one must be aware that positional errors already in the paper map will further accumulate, and one must be willing to accept these errors.
- There are two forms of digitizing: i) on-tablet and ii) on-screen manual digitizing.
- In on-tablet digitizing, the original map is fitted on a special surface (the tablet), while in on-screen digitizing, a scanned image of the map (or some other image) is shown on the computer screen.
- Digitizing involves converting hardcopy maps, plans, or other graphical sources into digital format.
- This process typically entails tracing geographic features manually from a printed map or image and entering the spatial data into a GIS system. It results in vector data representing point, line, or polygon features.

- **Scanning**

- An 'office' scanner illuminates a document and measures the intensity of the reflected light with a CCD array.
- The result is an image as a matrix of pixels, each of which holds an intensity value.
- Office scanners have a fixed maximum resolution, expressed as the highest number of pixels they can identify per inch.
- Scanning involves converting hardcopy maps or documents into digital images using a scanner.
- While scanning itself does not create GIS-ready data, the resulting digital images can be georeferenced or used as a base layer for further digitization.

9. What is satellite based positioning? Explain.**(Nov-19)****Ans:**

- **Satellite-based positioning**

- Satellites have allowed us to realize geocentric reference systems and increase the level of spatial accuracy substantially.
- They are critical tools in geodetic engineering for the maintenance of the ITRF.
- They also play a key role in mapping, surveying, and in a growing number of applications requiring positioning techniques.
- Nowadays, for fieldwork that includes spatial data acquisition, the use of satellite-based positioning is considered indispensable.
- Satellite-based positioning was developed and implemented to address military needs, somewhat analogously to the early development of the internet.
- Satellite-based positioning in GIS refers to the use of satellite systems to determine the geographic location of points on the Earth's surface.
- This technology enables accurate positioning, mapping, and navigation in GIS applications.
- The two primary satellite systems used for positioning are the Global Positioning System (GPS) and other similar satellite navigation systems.
- The technology is now widely available for civilians use. The requirements for the development of the positioning system were:

- i. Suitability for all kinds of military use: ground troops and vehicles, aircraft and
- ii. missiles, ships;
- iii. Requiring only low-cost equipment with low energy consumption at the receiver end;
- iv. Provision of results in real time for an unlimited number of users concurrently;

- **A satellite-based positioning system set-up involves implementation of three hardware segments:**

- i. The space segment, i.e. the satellites that orbit the Earth, and the radio signals that
- ii. they emit,
- iii. The control segment, i.e. the ground stations that monitor and maintain the space
- iv. segment components, and
- v. The user segment, i.e. the users with their hard and software to conduct positioning.

10. List the four issues in combining data from multiple sources. Explain any two of them. (Nov-19)

ANS:

Four issues in combining data from multiple sources:

A GIS project usually involves multiple data sets, so the next step addresses the issue of how these multiple sets relate to each other. There are four fundamental cases to be considered in the combination of data from different sources:

1. They may be about the same area, but differ in accuracy,
2. They may be about the same area, but differ in choice of representation,
3. They may be about adjacent areas and have to be merged into a single data set.
4. They may be about the same or adjacent areas but referenced in different coordinate systems.

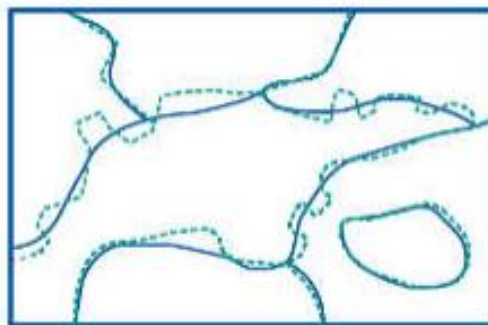
A range of other data preparation functions exist that support conversion or adjustment of data to format requirements that have been defined for data storage purposes. These include

Differences in representation

- When points need to be translated into rasters, we need to perform something known as point data transformation.
- Some advanced GIS applications require the possibility of representing the same geographic phenomenon in different ways. These are called multi representation systems.
- The commonality is that phenomena must sometimes be Multi-scale and viewed as points, and at other times as polygons.
- For example, a small-scale multi representation systems national road network analysis may represent villages as point objects, but a nation-wide urban population density study should regard all municipalities as represented by polygons.
- The links between various representations for the same object maintained by the system allows switching between them, and many fancy applications of their use seem possible.

Differences in accuracy:

Figure: The integration of two vector datasets, which represent the same phenomenon, may lead to sliver polygons



- In the above Figure the polygons of two digitized maps at different scales are overlaid.
- Due to scale differences in the sources, the resulting polygons do not perfectly coincide, and polygon boundaries cross each other.
- This causes small, artefact polygons in the overlay known as sliver polygons.
- If the map scales in- Sliver polygons varied differ significantly, the polygon boundaries of the large-scale map should probably take priority, but when the differences are slight, we need interactive techniques to resolve the issues.



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11. Write a note on GLONASS.**(Nov-19)****Ans**

- **GLONASS:**

- At present, two satellite-based positioning systems are operational (GPS and GLONASS), and a third is in the implementation phase (Galileo). Respectively, these are American, Russian and European systems.
- Any of these, but especially GPS and Galileo, will be improved over time, and will be augmented with new techniques.
- What GPS is to the US military, is GLONASS to the Russian military.
- The GLONASS space segment consists of nominally 24 satellites, organized in three orbital planes, with an inclination of 64.8° with the equator.
- Orbiting altitude is 19,130 km, with a period of revolution of 11 hours 16 min. GLONASS uses the PZ-90 as its reference system, and like GPS uses UTC as time reference, though with an offset for Russian daylight.
- GLONASS radio signals are somewhat similar to that of GPS, but differ in the details.
- Satellites use different identifier schemes, and their navigation message use other parameters.
- They also use different frequencies: GLONASS L1 is at approximately 1605 MHz (changes are underway), and L2 is at approximately 1248 MHz.
- Otherwise, the GLONASS system performance is rather comparable to that of GPS.
- Versions of GLONASS:
 - GLONASS - These satellites were launched in 1982 for the military and official organizations. They were intended to for weather, positioning, timing and velocity measurements.
 - GLONASS-M - These satellites were launched in 2003 to add second civil code, which is important for GIS mapping receivers.
 - GLONASS-K - These satellites were launched in 2011 to add third civil frequency. These are of 3 types - K1, K2 and KM.
 - GLONASS-K2 - These satellites will be launched after 2015 (currently in design phase).
 - GLONASS-KM - These satellites will be launched after 2025 (currently in research phase).

GLONASS Evolution

12. Explain the reference surface for mapping the Earth's surface. (Apr-23)**Ans:**

The surface of the Earth is anything but uniform. The oceans can be treated as reasonably uniform, but the surface or topography of the land masses exhibits large vertical variations between mountains and valleys. These variations make it impossible to approximate the shape of the Earth with any reasonably simple mathematical model. Two main reference surfaces have been established to approximate the shape of the Earth. One reference surface is called the Geoid, the other reference surface is the ellipsoid as shown in the figure below.

Imagine that the entire Earth's surface is covered by water. If ignored tidal and current effects on this 'global ocean', the resultant water surface is affected only by gravity. This has an effect on the shape of this surface because the direction of gravity- more commonly known as plumb line-is dependent on the mass distribution inside the Earth. Due to irregularities or mass anomalies in this distribution the 'global ocean' results in an undulated surface. This surface is called the Geoid. The plumb line through any surface point is always perpendicular to it.

The Geoid is used to describe heights. In order to establish the Geoid as reference for heights, the ocean's water level is registered at coastal places over several years using tide gauges (mareographs). Averaging the registrations largely eliminates variations of the sea level with time. The resulting water level represents an approximation to the Geoid and is called the mean sea level. The height of a point in Mumbai with respect to the tide gauge is measured using a technique known as geodetic levelling. The result of this process will be the height above local mean sea level for the Mumbai tidal gauge station point. The height determined with respect to a tide-gauge station is known as the orthometric height i.e. height H above the Geoid.

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13. Explain the working of GPS.**(Apr-23)****Ans:**

The Global Positioning System (GPS) plays a crucial role in GIS (Geographic Information System) by providing accurate and real-time positional information. The working of GPS in GIS involves the use of satellite signals to determine the precise geographic coordinates of a receiver. Here's a step-by-step explanation of how GPS works in GIS:

- **Satellite Constellation:**

Description: The GPS system consists of a constellation of satellites orbiting the Earth. As of my last knowledge update in January 2022, there are at least 24 GPS satellites in medium Earth orbit.

- **GPS Receiver:**

Description: A GPS receiver is a device that users carry, whether as standalone devices, integrated into smartphones, or part of GIS equipment. The receiver communicates with the GPS satellites to determine its location.

- **Satellite Signals:**

Transmission: Each GPS satellite continuously broadcasts signals that include information about the satellite's identity, its position in orbit, and the precise time the signal was transmitted.

Triangulation: The GPS receiver collects signals from multiple satellites simultaneously. To determine its location, the receiver triangulates its position based on the signals received from at least three or four satellites.

- **Distance Calculation:**

Trilateration: The GPS receiver calculates its distance from each satellite by measuring the time it takes for the signals to travel from the satellites to the receiver. Knowing the speed of light, the receiver can convert the time delay into distance.

- **Position Calculation:**

Trilateration Algorithm: Using the distances calculated from the signals received from multiple satellites, the GPS receiver employs a trilateration algorithm to determine its precise three-dimensional geographic coordinates (latitude, longitude, and altitude).

- **Accuracy Improvement:**

Correction Techniques: To enhance accuracy, advanced GPS receivers may use correction techniques. Differential GPS (DGPS) involves receiving correction signals from a stationary reference station to compensate for atmospheric and signal delays, improving accuracy to centimeter-level.

- **Integration with GIS:**

Spatial Data Integration: The determined GPS coordinates are integrated into the GIS system, allowing users to visualize and analyze real-world spatial data. These coordinates become reference points for mapping, navigation, or any GIS-related activity.

- **Real-Time Updates:**

Dynamic Positioning: For dynamic applications, such as vehicle navigation or field data collection, the GPS receiver continuously updates its position in real-time. This enables users to track movement and capture spatial data dynamically.

In summary, the GPS system in GIS works by receiving signals from multiple satellites, calculating distances based on signal travel times, and using trilateration algorithms to determine precise geographic coordinates. The integration of GPS data into GIS allows for accurate mapping, navigation, and analysis of spatial information.



14. Write a short note on vectorization.**(Apr-23)****Ans**

- Vectorization produces a vector data set from a raster.
- Another form of vectorization takes place when we want to identify features or patterns in remotely sensed imagery.
- The keywords here are feature extraction and pattern recognition, which are dealt with Remote Sensing.
- Rasterization is a process of converting vector data sets to raster data when some or all of the subsequent spatial data analysis is to be carried out on raster data.
- It involves assigning point, line and polygon attribute values to raster cells that overlap with the respective point, line or polygon. Data Entry and Preparation 103
- A cell size which is too large may result in cells that cover parts of multiple vector features.
- Rasterization itself could be seen as a 'backwards step':
 - Firstly, raster boundaries are only an approximation of the objects' original boundary.
 - Secondly, the original 'objects' can no longer be treated as such, as they have lost their topological properties.
- An alternative to rasterization is to not perform it during the data preparation phase, but to use GIS rasterization functions on- the-fly, that is when the computations call for it.
- This allows keeping the vector data and generating raster data from them when needed. Obviously, the issue of performance trade-off must be looked into.
- Vectorization in GIS refers to the process of converting raster data (imagery or scanned maps) into vector data. Vector data represent spatial features using points, lines, and polygons, which offer distinct advantages for spatial analysis and mapping.
- The vectorization process involves identifying and delineating features in raster images and creating corresponding vector representations.
- vectorization is a fundamental process in GIS that converts raster data into vector representations. This conversion enhances the capabilities of GIS by providing accurate, precise, and easily analyzable spatial datasets.

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15. What is Interpolation? Explain interpolation of continuous Data.**(Apr-23)****Ans:****Interpolation**

- We may want to transform our points into other representations in order to facilitate interpretation and/or integration with other data.
- Examples include defining homogeneous areas (polygons) from our point data, or deriving contour lines.
- This is generally referred to as interpolation,
- The calculation of an Interpolation value from 'surrounding' observations.
- The principle of spatial autocorrelation plays a central part in the process of interpolation.

➤ Interpolating continuous data

There are many continuous geographic field elevation, temperature and groundwater salinity are just a few examples. Commonly, continuous fields are represented as rasters, and we will almost by default assume that they are.

The main alternative for continuous field representation is a polyline vector layer, in which the lines are isolines.

The aim is to use measurements to obtain a representation of the entire field using point samples.

In this section we outline four techniques to do so:

- 1 Trend surface fitting using regression,
- 2 Triangulation,
- 3 Spatial moving averages using inverse distance weighting,
- 4 Kriging

1 Trend surface fitting using regression,

- A surface interpolation method that fits a polynomial surface by leastsquares regression through the sample data points.
- This method results in a surface that minimizes the variance of the surface in relation to the input values.
- The resulting surface rarely goes through the sample data points.

2 Triangulation,

- Triangulation refers to the use of multiple methods or data sources in qualitative research to develop a comprehensive understanding of phenomena (Patton, 1999).
- Triangulation also has been viewed as a qualitative research strategy to test validity through the convergence of information from different sources.

3 Spatial moving averages using inverse distance weighting,

- Inverse distance weighted (IDW) interpolation determines cell values using a linearly weighted combination of a set of sample points.
- The weight is a function of inverse distance. The surface being interpolated should be that of a locationally dependent variable.

4 Kriging.

- Kriging is a multistep process; it includes exploratory statistical analysis of the data, variogram modeling, creating the surface, and (optionally) exploring a variance surface.
- Kriging is most appropriate when you know there is a spatially correlated distance or directional bias in the data.
- Kriging predicts the value of a function at a given point by computing a weighted average of the known values of the function in the neighborhood of the point.
- The method is closely related to regression analysis.

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16. Write a short note on Datum.**ANS**

A datum is a set of reference points on the earth's surface against which position measurements are made, and (often) an associated model of the shape of the earth (reference ellipsoid) to define a geographic coordinate system.

In GIS (Geographic Information System), a datum is a reference system used to define the location of spatial data on the Earth's surface. It serves as the foundation for establishing coordinate systems and ensuring accurate geospatial positioning. A datum comprises a reference ellipsoid, which represents the shape of the Earth, and a reference point known as the origin.

- **Importance of Datum in GIS:**

- 1 Coordinate Consistency:**

Different countries and regions historically used their own datums, resulting in inconsistencies in spatial data. Standardized datums, such as WGS84, promote global coordination and consistency in GIS applications.

- 2 Accurate Positioning:**

The choice of datum directly influences the accuracy of spatial positioning. GIS data must be referenced to a specific datum to ensure precise and reliable location information.

- 3 Interoperability:**

Datum standardization facilitates data sharing and interoperability across different GIS platforms and systems. It ensures that spatial datasets align seamlessly when integrated or exchanged between organizations.

- 4 Map Projection Alignment:**

Datum compatibility is crucial when working with map projections. Maps and GIS data must be referenced to the same datum to avoid distortions and misalignments when using various map projections.

- 5 Geodetic Transformations:**

In some cases, GIS data may need to be transformed from one datum to another to ensure alignment. Geodetic transformations are applied to adjust coordinates when moving between different datum reference systems.

6 Global Positioning System (GPS):

GPS receivers use specific datums to provide accurate positioning information. GIS applications that incorporate GPS data need to ensure that the datum used in the GIS matches the datum used by the GPS receiver.

In conclusion, datum is a critical concept in GIS, providing the reference framework necessary for accurate geospatial data representation and analysis. Standardizing datums promotes consistency, interoperability, and accurate positioning in the field of geographic information systems.

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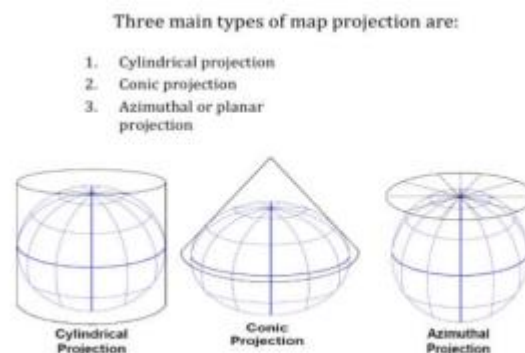
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17. What is Map Projection? List and explain types of Map Projections.**OR****List and explain commonly used map projections.****ANS**

A map projection is one of many methods used to represent the 3- dimensional surface of the earth or other round body on a 2-dimensional plane in cartography (mapmaking).

The most common list of map projections include:

**1) Equal area projections**

- These projections preserve the area of specific features.
- These projections distort shape, angle, and scale.
- The Albers Equal Area Conic projection is an example of an equal area projection.

2) Conformal projections

- These projections preserve local shape for small areas.
- These projections preserve individual angles to describe spatial relationships by showing perpendicular graticule lines that intersect at 90-degree angles on the map.
- All the angles are preserved; however, the area of the map is distorted.
- The Mercator and Lambert Conformal Conic projections are examples of conformal projections.

3) Equidistant projections

- These projections preserve the distances between certain points by maintaining the scale of a given data set.
- Some of the distances will be true distances, which are the same distances at the same scale as the globe.
- If you go outside the data set, the scale will become more distorted.
- The Sinusoidal projection and the Equidistant Conic projection are examples of equidistant projections.

4) True-direction or azimuthal projections

- These projections preserve the direction from one point to all other points by maintaining some of the great circle arcs.
- These projections give the directions or azimuths of all points on the map correctly with respect to the center.
- Azimuthal maps can be combined with equal area, conformal, and equidistant projections.
- The Lambert Equal Area Azimuthal projection and the Azimuthal Equidistant projection are examples of azimuthal projections.



18. Explain different Coordinate systems used in GIS.

Ans:

A coordinate system is a method for identifying the location of a point on the earth. Most coordinate systems use two numbers, a coordinate, to identify the location of a point. Different kinds of coordinate systems are used to position data in space. Spatial (or global) coordinate systems are used to locate data either on the Earth's surface in a 3D space, or on the Earth's reference surface in a 2D space. The geographic coordinate system in 2D and 3D space and the geocentric coordinate system, also known as the 3D Cartesian coordinate system. Planar coordinate systems on the other hand are used to locate data on the flat surface of the map in a 2D space.

1. 2D Geographic coordinates

The most widely used global coordinate system consists of lines of geographic latitude (ϕ or ϕ) and longitude (λ or λ). Lines of equal latitude are called parallels. They form circles on the surface of the ellipsoid. Lines of equal longitude are called meridians and they form ellipses (meridian ellipses) on the ellipsoid. The latitude (ϕ) of a point P is the angle between the ellipsoidal normal through P' and the equatorial plane. Latitude is zero on the equator ($\phi=0^\circ$), and increases towards the two poles to maximum values of $+90^\circ$ (N 90°) at the North Pole and -90° (S 90°) at the South Pole. The longitude (λ) is the angle between the meridian ellipse which passes through Greenwich and the meridian ellipse containing the point in question. It is measured in the equatorial plane from the meridian of Greenwich ($\lambda=0^\circ$) either eastwards through $\lambda=+180^\circ$ (E 180°) or westwards through $\lambda=-180^\circ$ (W 180°).

2. 3D Geographic coordinates (ϕ, λ, h)

3D geographic coordinates (ϕ, λ, h) are obtained by introducing the ellipsoidal height h to the system. The ellipsoidal height (h) of a point is the vertical distance of the point in question above the ellipsoid. It is measured in distance units along the ellipsoidal normal from the point to the ellipsoid surface. 3D geographic coordinates can be used to define a position on the surface of the Earth (point P).

3. 3D Geocentric coordinates (X,Y,Z)

An alternative method of defining a 3D position on the surface of the Earth is by means of geocentric coordinates (X,Y,Z) also known as 3D Cartesian coordinates. The system has its origin at the mass-centre of the Earth with the X and Y axes in the plane of the equator. The X-axis passes through the meridian of Greenwich, and the Z-axis coincides with the Earth's axis of rotation. The three axes are mutually orthogonal and form a right-handed system. Geocentric coordinates can be used to define a position on the surface of the Earth (point P in Figure). It should be noted that the rotational axis of the earth changes its position over time (referred to as polar motion). To compensate for this, the mean position of the pole in the year 19 (based

on observations between 1900 and 1905) has been used to define the so-called Conventional International Origin (CIO).

4. 2D Cartesian coordinates (X,Y)

A flat map has only two dimensions: width (left to right) and length (bottom to top). Transforming the three dimensional Earth into a two-dimensional map is subject of map projections and coordinate transformation. Like in several other cartographic applications, two-dimensional Cartesian coordinates (x, y), also known as planar rectangular coordinates, are used to describe the location of any point unambiguously. The two coordinates x and y for point P, specify any location P on the map.

5. 2D Polar coordinates (a, d)

Polar coordinate is the distance "d" from the origin to the point concerned and the angle a between a fixed (or zero) direction and the direction to the point. The angle a is called azimuth or bearing and is measured in a clockwise direction. It is given in angular units while the distance d is expressed in length units. Bearings are always related to a fixed direction (initial bearing) or a datum line. In principle, this reference line can be chosen freely. However, in practice three different directions are widely used: True North, Grid North and Magnetic North. The corresponding bearings are called: true (or geodetic) bearing, grid bearing and magnetic (or compass) bearing.

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19. Explain Spatial framework for mapping locations.

Ans:

A spatial framework for mapping locations provides a systematic structure for representing and organizing geographic information. It serves as the foundation for creating maps, conducting spatial analyses, and understanding the relationships between different locations on the Earth's surface. The spatial framework involves several key components:

➤ **Coordinate System:**

Definition: A coordinate system is a reference system used to represent locations on the Earth's surface. It involves a grid of lines (latitude and longitude or easting and northing) that assigns unique numerical coordinates to each point.

Role: Coordinates serve as a fundamental aspect of the spatial framework, allowing precise location identification and referencing.

➤ **Datum:**

Definition: A datum defines the reference ellipsoid and origin point for a coordinate system. It ensures that geographic data aligns accurately with the Earth's surface.

Role: Datum provides a standardized reference for mapping and ensures consistency in spatial representation.

➤ **Map Projection:**

Definition: A map projection is a mathematical transformation that converts the three-dimensional Earth's surface onto a two-dimensional map. Different map projections have distinct properties, and their selection depends on the purpose of the map.

Role: Map projections help represent the Earth's curved surface on a flat map while minimizing distortions in area, shape, distance, or direction.

➤ **Scale:**

Definition: Scale refers to the relationship between distances on a map and the corresponding distances on the Earth's surface. It can be expressed as a ratio, representative fraction, or verbal scale.

Role: Scale is crucial for understanding the level of detail, size, and extent of geographic features on a map.

➤ **Topological Relationships:**

Definition: Topology defines the spatial relationships between geographic features, such as connectivity, adjacency, or containment. It helps establish how features interact with each other.

Role: Topology ensures the integrity of spatial data and facilitates spatial analysis, query, and overlay operations.

➤ **Spatial Reference Systems:**

Definition: Spatial reference systems encompass coordinate systems, datums, and map projections, providing a comprehensive framework for spatial data representation.

Common spatial reference systems include Geographic Coordinate Systems (GCS) and Projected Coordinate Systems (PCS).

Role: Spatial reference systems establish a consistent framework for locating and analyzing features in a GIS environment.

➤ **Geographic Information Systems (GIS):**

Definition: GIS is a technology that combines spatial data (maps and coordinates) with attribute data (information linked to geographic features) to analyze, visualize, and interpret spatial patterns and relationships.

Role: GIS integrates the spatial framework into a dynamic environment, allowing users to store, analyze, and visualize geographic information.

➤ **Metadata:**

Definition: Metadata provides information about the characteristics, quality, and context of geographic data. It includes details about the source, accuracy, date, and other relevant attributes.

Role: Metadata enhances the usability and understanding of spatial data, ensuring proper interpretation and application.

Together, these components form a spatial framework that supports accurate, consistent, and meaningful representation of locations in mapping and GIS activities. This framework is essential for creating reliable and effective spatial data products for various applications, ranging from navigation to environmental analysis.

20. Distinguish between primary and secondary data and give examples of each. In what circumstances is this distinction difficult to maintain?

Ans:

❖ **Primary Data:**

➤ **Definition:**

Primary data refers to original data collected directly from its source for a specific purpose or research objective. It is firsthand information obtained through surveys, fieldwork, experiments, or direct observation.

➤ **Characteristics:**

- Collected for a specific purpose.
- Current and specific to the research question.
- More control over data quality and relevance.

➤ **Examples in GIS:**

- Conducting a field survey to collect GPS coordinates and attribute information about the location of tree species in a forest.
- Using a GPS device to record real-time coordinates during a wildlife tracking expedition.

❖ **Secondary Data:**

➤ **Definition:**

Secondary data refers to existing data that has been collected for a purpose other than the current research or application. It is data that someone else has collected, processed, and made available for reuse.

➤ **Characteristics:**

- Collected for a different purpose or project.
- Can be historical or more general in nature.
- Often less control over data quality or relevance.

➤ **Examples in GIS:**

- Using population data obtained from a national census for a regional planning project.
- Utilizing satellite imagery collected by a space agency for land cover classification.

In summary, while primary data is collected directly for a specific purpose, and secondary data already exists for a different purpose, GIS projects often require the integration and careful management of both types. Maintaining the distinction can be challenging due to the dynamic nature of GIS work, which involves data integration, transformation, updates, and quality assurance.

21. Rasterization of vector data is sometimes required in data preparation.**What reasons may exist for this?****Ans:**

Rasterization of vector data is a process in GIS where vector data, represented by points, lines, and polygons, is converted into a raster format, where the data is represented as a grid of cells or pixels. This transformation is often necessary for various reasons in data preparation. Here are some common reasons for rasterizing vector data in GIS:

- **Spatial Analysis:**

Reason: Raster data is well-suited for certain types of spatial analysis, such as overlay operations, terrain modeling, and proximity analysis. Raster datasets allow for the application of mathematical operations across a continuous surface, which is beneficial for certain analytical tasks.

Example: Calculating slope, aspect, or terrain ruggedness, where the output is better represented in a raster format.

- **Interpolation and Surface Modeling:**

Reason: Rasterization is often used in interpolation methods to create continuous surfaces from point data. This is valuable for modeling spatial phenomena across an area, especially when working with elevation, temperature, or other continuous variables.

Example: Creating a raster surface representing temperature distribution based on temperature measurements taken at specific locations.

- **Data Integration:**

Reason: Raster data is more compatible with other raster datasets, making integration and overlay analysis more straightforward. Rasterized vector data can be easily combined with other raster layers for comprehensive spatial analysis.

Example: Overlaying land cover (raster) with administrative boundaries (vector) to derive land cover information within specific administrative units.

- **Cartographic Representation:**

Reason: Raster datasets are often used for cartographic representation when continuous or smoothly varying features need to be visualized. This is especially true for environmental modeling or thematic mapping.

Example: Creating a raster map to represent vegetation types, where the continuous nature of raster cells can better represent gradual changes in vegetation.

- **Data Simplification:**

Reason: Rasterization can be used as a form of data simplification, especially when dealing with complex vector datasets. This simplification can make the data more manageable and reduce processing time.

Example: Converting detailed vector road networks into a raster format when the primary focus is on broad-scale network analysis rather than detailed vector representation.

- **Efficient Storage and Processing:**

Reason: Raster datasets are generally more efficient in terms of storage and processing for certain operations, particularly when dealing with large-scale spatial datasets.

Example: Rasterizing vector land cover data to facilitate faster processing of algorithms like least-cost path analysis.

- **Compatibility with Remote Sensing Data:**

Reason: Raster data is commonly used in conjunction with remote sensing data, which is often collected in raster format. Converting vector data to raster allows for seamless integration with remote sensing datasets.

Example: Combining vector land parcel data with satellite imagery for land cover classification.

In summary, rasterization of vector data in GIS is a versatile process that serves various purposes, including spatial analysis, interpolation, integration, cartographic representation, and simplification, depending on the specific requirements of the GIS project. The choice to rasterize vector data is often driven by the nature of the analysis and the desired output format for effective data preparation and subsequent spatial modeling.



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22. List the four issues in obtaining data from multiple sources. Explain any two of them.

Ans:

Obtaining data from multiple sources in GIS can pose several challenges that may impact the quality, consistency, and interoperability of the datasets. Here are four common issues, along with explanations for two of them:

1) Spatial Inconsistency:

Explanation: Spatial inconsistency arises when datasets from different sources do not align correctly in terms of coordinate systems, projections, or spatial referencing.

Mismatched spatial references can lead to misalignments, inaccuracies, and difficulties in combining or overlaying datasets.

Example: Combining a satellite image referenced in one coordinate system with vector data collected in a different coordinate system without proper transformation may result in spatial inconsistencies, leading to misalignment and errors in spatial analysis.

2) Attribute Inconsistency:

Explanation: Attribute inconsistency occurs when there are variations in the structure, format, or semantics of attribute data across different datasets. Incompatible attribute data may hinder the meaningful integration and analysis of information.

Example: Combining datasets where one records population using whole numbers and another uses decimals for the same attribute can lead to attribute inconsistencies. The variation in data formats may result in challenges during analysis and interpretation.

3) Temporal Inconsistency:

Explanation: Temporal inconsistency arises when datasets have different temporal resolutions, time formats, or reference points. Merging data with temporal inconsistencies can complicate time-based analyses and result in erroneous conclusions.

Example: Integrating weather data recorded at hourly intervals with climate data recorded annually may lead to challenges in performing temporal analyses without proper alignment or interpolation. Temporal inconsistencies can affect the accuracy of time-dependent analyses.

4) Data Quality Disparities:

Explanation: Data quality disparities refer to differences in the accuracy, precision, completeness, or currency of data from various sources. Combining datasets with varying data quality can compromise the reliability of analyses and decision-making.

Example: Integrating high-resolution, accurately surveyed data with lower-quality, less accurate data may introduce errors into the integrated dataset. Users need to be aware of the potential impact of data quality differences on the results.

Addressing these issues requires careful consideration of data standards, preprocessing steps, and the use of transformation methods to ensure compatibility and consistency across multiple datasets. Metadata documentation, standardized formats, and spatial or attribute matching techniques are commonly employed to overcome these challenges in GIS data integration.



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